

1. Check if a Directory is Available or Not

Step 1: Create a Script File

Create a new Bash script using the vi editor.

```
vi check_hari.sh
```

Step 2: Press Insert Mode:

```
#!/bin/bash
DIR="/home/ec2-user/test"
if [ -d "$DIR" ]
then
    echo "Directory exists"
else
    echo "Directory does not exists."
fi
```

Step 2: Give Execute Permission

```
chmod 755 check_hari.sh
```

Step 3: Execute the Script

```
bash check_hari.sh
```

Output

If the directory exists:

Directory exists.

If the directory does not exist:

Directory does not exist.

```
[root@ip-172-31-32-82 ec2-user]# vi check_hari.sh
[root@ip-172-31-32-82 ec2-user]# chmod 755 check_hari.sh
[root@ip-172-31-32-82 ec2-user]# bash check_hari.sh
Directory does not exist.
```

2. Create Multiple Files

Step 1: Create a Script File:

Create a new Bash script using the vi editor.

```
vi create_multiple_files.sh
```

Step 2: Enter Insert Mode:

```
root@ip-172-31-32-82:/home/ec2-user
#!/bin/bash
for i in {5..10}
do
    touch file$i.txt
done
echo "5 files created successfully."
```

Step 4: Give Execute Permission

Run the following command to make the script executable:

```
chmod 755 create_multiple_files.sh
```

Step 5: Execute the Script:

```
bash create_multiple_files.sh
```

Step 6: Verify the Files

Use the following command to list the files created:

ls

```
[root@ip-172-31-32-82 ec2-user]# vi create_multiple_files.sh
[root@ip-172-31-32-82 ec2-user]# chmod 755 create_multiple_files.sh
[root@ip-172-31-32-82 ec2-user]# bash create_multiple_files.sh
5 files created successfully.
[root@ip-172-31-32-82 ec2-user]# ls -l
-rw-r--r-- 1 root root 4096 Jul 5 2026 a
-rw-r--r-- 1 root root 4096 Jul 5 2026 an
-rw-r--r-- 1 root root 4096 Jul 5 2026 b
-rw-r--r-- 1 root root 4096 Jul 5 2026 c
-rwxr-xr-x 1 root root 4096 Jul 5 2026 check_hari.sh
-rwxr-xr-x 1 root root 4096 Jul 5 2026 create_multiple_files.sh
-rw-r--r-- 1 root root 4096 Jul 5 2026 file10.txt
-rw-r--r-- 1 root root 4096 Jul 5 2026 file5.txt
-rw-r--r-- 1 root root 4096 Jul 5 2026 file6.txt
-rw-r--r-- 1 root root 4096 Jul 5 2026 file7.txt
-rw-r--r-- 1 root root 4096 Jul 5 2026 file8.txt
-rw-r--r-- 1 root root 4096 Jul 5 2026 file9.txt
-rwxr-xr-x 1 root root 4096 Jul 5 2026 hari.sh
-rwxr-xr-x 1 root root 4096 Jul 5 2026 input.sh
-rwxr-xr-x 1 root root 4096 Jul 5 2026 output.sh
-rwxr-xr-x 1 root root 4096 Jul 5 2026 readinput.sh
-rwxr-xr-x 1 root root 4096 Jul 5 2026 run
-rwxr-xr-x 1 root root 4096 Jul 5 2026 srihari.sh
-rwxr-xr-x 1 root root 4096 Jul 5 2026 task_completed
-rwxr-xr-x 1 root root 4096 Jul 5 2026 test
-rwxr-xr-x 1 root root 4096 Jul 5 2026 variables.sh
```

3.Create a bash script to take a backup of a directory.

Step 1: Create the Script File

```
vi directory.sh
```

```
root@ip-172-31-18-95:/home/ec2-user_backup
#!/bin/bash
#purpose - To create backup of a directory
#owner - Srihari
#Modification DATE - 7/5/2026
#!/bin/bash

cp -r /home/ec2-user /home/ec2-user_backup
echo "Backup created successfully"
```

Step 2: Verify the Output

```
root@ip-172-31-18-95:/home/ec2-user_backup
[root@ip-172-31-18-95 ec2-user]# vi directory.bash
[root@ip-172-31-18-95 ec2-user]# chmod 755 directory.bash
[root@ip-172-31-18-95 ec2-user]# bash directory.bash
Backup created successfully
```

4.Create a bash script to install Nginx on an EC2 server.

```
root@ip-172-31-18-95:/home/ec2-user
#!/bin/bash
#purpose - to install nginx on ec2 server.
#owner - Srihari
#Modification date - 7/5/2026

#!/bin/bash

sudo yum update -y

sudo yum install nginx -y

sudo systemctl start nginx

sudo systemctl enable nginx

sudo systemctl status nginx

echo "Nginx installed successfully"
```

```
root@ip-172-31-18-95:/home/ec2-user
[root@ip-172-31-18-95 ec2-user]# vi nginx.bash
[root@ip-172-31-18-95 ec2-user]# chmod 755 mnginx.bash
chmod: cannot access 'mnginx.bash': No such file or directory
[root@ip-172-31-18-95 ec2-user]# chmod 755 nginx.bash
[root@ip-172-31-18-95 ec2-user]# bash nginx.bash
Amazon Linux 2023 Kernel Livepatch repository      406 kB/s | 55 kB      00:00
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:00:03 ago on Sun Jul  5 09:05:55 2026.
Dependencies resolved.
=====
Package                Architecture  Version                                Repository      Size
=====
Installing:
nginx                  x86_64       1:1.30.2-1.amzn2023.0.1               amazonlinux     34 k
Installing dependencies:
generic-logos-httpd    noarch       18.0.0-12.amzn2023.0.3               amazonlinux     19 k
gperftools-libs        x86_64       2.9.1-1.amzn2023.0.3                 amazonlinux    308 k
libunwind              x86_64       1.4.0-5.amzn2023.0.3                 amazonlinux     66 k
=====
Nginx installed successfully
[root@ip-172-31-18-95 ec2-user]# sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-07-05 09:06:00 UTC; 5min ago
     Main PID: 3839 (nginx)
       Tasks: 3 (limit: 1059)
      Memory: 3.4M
         CPU: 77ms
       CGroup: /system.slice/nginx.service
              └─3839 "nginx: master process /usr/sbin/nginx"
                 └─3840 "nginx: worker process"
                    └─3841 "nginx: worker process"
```

Not secure 51.20.66.149

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

5.Create a bash script to install Apache Tomcat on an EC2 server.

```
root@ip-172-31-18-95:/home/ec2-user
#!/bin/bash
#purpose - To install apache tomcat on ec2 server
#owner - Srihari
#Modification date - 7/5/2026

sudo yum update -y

sudo yum install java-11-amazon-corretto -y

cd /opt

sudo wget https://d1cdn.apache.org/tomcat/tomcat-9/v9.0.119/bin/apache-tomcat-9.0.119.tar.gz

sudo tar -xvzf apache-tomcat-9.0.119.tar.gz

sudo mv apache-tomcat-9.0.119 tomcat

echo "Tomcat installed and started successfully"
```

```
root@ip-172-31-18-95:/home/ec2-user
[root@ip-172-31-18-95 ec2-user]# vi tomcat.bash
[root@ip-172-31-18-95 ec2-user]# vi tomcat.bash
[root@ip-172-31-18-95 ec2-user]# bash tomcat.bash
Last metadata expiration check: 0:18:57 ago on Sun Jul  5 09:05:55 2026.
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:18:58 ago on Sun Jul  5 09:05:55 2026.
Dependencies resolved.

=====
Package                        Arch      Version                               Repository      Size
=====
Installing:
java-11-amazon-corretto        x86_64    1:11.0.31+11-1.amzn2023              amazonlinux     196 k
apache-tomcat-9.0.119/bin/shutdown.sh
apache-tomcat-9.0.119/bin/startup.sh
apache-tomcat-9.0.119/bin/tool-wrapper.sh
apache-tomcat-9.0.119/bin/version.sh
Tomcat installed and started successfully
```

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Apache Tomcat/9.0.119

If you're seeing this, you've successfully installed Tomcat. Congratulations!

 Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

Server Status
Manager App
Host Manager

6. Create a bash script to check if the Nginx service is running, if not running then script should start the service.

```
root@ip-172-31-18-95:/home/ec2-user
#!/bin/bash
#purpose : nginx service is running
#owner: Srihari
#date: 7/5/2026
#check if nginx service is running

if systemctl is-active --quit nginx
then
    echo "Nginx service is already running"
else
    echo "Nginx service is not running"

    # Start Nginx service
    systemctl start nginx
    systemctl status nginx

    echo "Nginx service started successfully."
fi

~

[root@ip-172-31-18-95 ec2-user]# vi nginx_check.sh
[root@ip-172-31-18-95 ec2-user]# bash nginx_check.sh
systemctl: unrecognized option '--quit'
Nginx service is not running
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-07-05 09:54:46 UTC; 1min 29s ago
     Process: 27837 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 27838 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 27839 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
    Main PID: 27840 (nginx)
      Tasks: 3 (limit: 1059)
     Memory: 3.6M
        CPU: 57ms
    CGroup: /system.slice/nginx.service
            └─27840 "nginx: master process /usr/sbin/nginx"
              └─27841 "nginx: worker process"
                └─27842 "nginx: worker process"

Jul 05 09:54:46 ip-172-31-18-95.eu-north-1.compute.internal systemd[1]: Starting nginx.service - Th>
Jul 05 09:54:46 ip-172-31-18-95.eu-north-1.compute.internal nginx[27838]: nginx: the configuration >
Jul 05 09:54:46 ip-172-31-18-95.eu-north-1.compute.internal nginx[27838]: configuration file>
Jul 05 09:54:46 ip-172-31-18-95.eu-north-1.compute.internal systemd[1]: Started nginx.service - The>
Nginx service started successfully.
```

7.Create a bash script for a calculator.

```
root@ip-172-31-18-95:/home/ec2-user
#!/bin/bash
#Purpose - - Bash script for calculator.
#Owner - Srihari
# Modification Date - 7/05/2026
#!/bin/bash

a=10
b=2

echo "BASH CALCULATOR"
echo "-----"
echo "Num1 = $a"
echo "Num2 = $b"
echo "-----"
echo "Addition      : $a + $b = $((a + b))"
echo "Subtraction   : $a - $b = $((a - b))"
echo "Multiplication : $a * $b = $((a * b))"
echo "Division       : $a / $b = $((a / b))"

echo "Modulus        : $a % $b = $((a % b))"

echo "power          : $a ^ $b = $(echo "$a ^ $b" | bc)"

echo "-----"
```

```
[root@ip-172-31-18-95 ec2-user]# vi cal.sh
[root@ip-172-31-18-95 ec2-user]# chmod 755 cal.sh
[root@ip-172-31-18-95 ec2-user]# bash cal.sh
BASH CALCULATOR
-----
Num1 = 10
Num2 = 2
-----
Addition      : 10 + 2 = 12
Subtraction   : 10 - 2 = 8
Multiplication : 10 * 2 = 20
Division       : 10 / 2 = 5
Modulus        : 10 % 2 = 0
power          : 10 ^ 2 = 100
-----
[root@ip-172-31-18-95 ec2-user]# |
```

8. Create a bash script to check if a directory exists, if not then create a directory.

```
#!/bin/bash
# Purpose          - Bash script check if a directory exists, if not then create a directory..
# Owner            - Srihari
# Modification Date - 7/05/2026
# !/bin/bash

DIR="/tmp/SrihariProject"

echo "=====
echo "  DIRECTORY CHECK & CREATE"
echo "=====
echo "Directory = $DIR"
echo "-----"

if [ -d "$DIR" ]; then
    echo "Status : Already Exists"
else
    echo "Status : Not Found"
    mkdir -p "$DIR"
    echo "Action : Created Successfully"
fi

echo "-----"
ls -ld "$DIR"
echo "=====
```

```
[root@ip-172-31-18-95 ec2-user]# vi check.sh
[root@ip-172-31-18-95 ec2-user]# chmod 755 check.sh
[root@ip-172-31-18-95 ec2-user]# bash check.sh
=====
  DIRECTORY CHECK & CREATE
=====
Directory = /tmp/SrihariProject
-----
Status : Not Found
Action : Created Successfully
-----
drwxr-xr-x. 2 root root 40 Jul  5 14:30 /tmp/SrihariProject
=====
[root@ip-172-31-18-95 ec2-user]#
```

9. Create a bash script to delete the last 3 lines of a file.

```
#!/bin/bash
# Purpose           - Bash script to delete the last 3 lines of a file..
# Owner            - Srihari
# Modification Date - 7/05/2026
# !/bin/bash

FILE="/tmp/myfile.txt"

# Create sample file
cat > "$FILE" << 'LINES'
Line 1: Server started
Line 2: User logged in
Line 3: Backup completed
Line 4: ERROR: Timeout
Line 5: ERROR: Retry failed
Line 6: ERROR: Service down
LINES

echo "=====
echo "  DELETE LAST 3 LINES"
echo "=====
echo "File = $FILE"
echo "-----"
echo "BEFORE:"
cat -n "$FILE"
echo "-----"

TOTAL=$(wc -l < "$FILE")
cp "$FILE" "${FILE}.bak"
head -n $(( TOTAL - 3 )) "$FILE" > "${FILE}.tmp" && mv "${FILE}.tmp" "$FILE"

echo "AFTER:"
cat -n "$FILE"
echo "-----"
echo "Lines Before : $TOTAL"
echo "Lines After  : $(wc -l < "$FILE")"
echo "Backup File   : ${FILE}.bak"
echo "=====

[root@ip-172-31-18-95 ec2-user]# vi check.sh
[root@ip-172-31-18-95 ec2-user]# chmod 755 check.sh
[root@ip-172-31-18-95 ec2-user]# ./check.sh
=====
DELETE LAST 3 LINES
=====
File = /tmp/myfile.txt
-----
BEFORE:
  1 Line 1: Server started
  2 Line 2: User logged in
  3 Line 3: Backup completed
  4 Line 4: ERROR: Timeout
  5 Line 5: ERROR: Retry failed
  6 Line 6: ERROR: Service down
-----
AFTER:
  1 Line 1: Server started
  2 Line 2: User logged in
  3 Line 3: Backup completed
-----
Lines Before : 6
Lines After  : 3
Backup File   : /tmp/myfile.txt.bak
=====
[root@ip-172-31-18-95 ec2-user]# |
```

10. Bash script to monitor cpu and if it is more than 80% then send email notification.

```
#!/bin/bash
#Purpose - Monitor cpu.
#Owner - Srihari
#Modification Date - 7/05/2026
#!/bin/bash
#!/bin/bash

THRESHOLD=80
EMAIL="admin@example.com"

echo "=====
echo "      CPU MONITOR SCRIPT"
echo "=====
echo "Threshold : ${THRESHOLD}%"
echo "Email      : $EMAIL"
echo "Timestamp  : $(date '+%Y-%m-%d %H:%M:%S')"
echo "Hostname   : $(hostname)"
echo "-----"

# Get CPU usage
CPU_IDLE=$(top -bn1 | grep "Cpu(s)" | awk '{for(i=1;i<=NF;i++) if($i~/id/){val=$(i-1); gsub(/%/,"",val)}; print val; exit}}')
CPU_USAGE=$(echo "100 - $CPU_IDLE" | bc | awk '{printf "%.1f", $1}')

echo "CPU Idle   : ${CPU_IDLE}%"
echo "CPU Usage  : ${CPU_USAGE}%"
echo "-----"

echo "Top 5 CPU Processes:"
ps -eo pid,comm,%cpu --sort=-%cpu | head -5
```

```
[root@ip-172-31-18-95 ec2-user]# vi monitor_cpu.sh
[root@ip-172-31-18-95 ec2-user]# chmod 755 monitor_cpu.sh
[root@ip-172-31-18-95 ec2-user]# bash monitor_cpu.sh
=====
      CPU MONITOR SCRIPT
=====
Threshold : 80%
Email      : admin@example.com
Timestamp  : 2026-07-05 14:52:24
Hostname   : ip-172-31-18-95.eu-north-1.compute.internal
-----
(standard_in) 1: syntax error
CPU Idle   : ni,100.0%
CPU Usage  : %
-----
Top 5 CPU Processes:
  PID COMMAND           %CPU
   1 systemd             0.0
   2 kthreadd             0.0
   3 pool_workqueue_     0.0
   4 kworker/R-rcu_g     0.0
   5 kworker/R-sync_     0.0
[root@ip-172-31-18-95 ec2-user]#
```

11. Bash script to monitor disk space and if it is more than 80% then send email notification.

```
#!/bin/bash

THRESHOLD=80
EMAIL="admin@example.com"

DISK_USAGE=$(df -h / | awk 'NR==2 {print $5}' | tr -d '%')

echo "=====|
echo "    DISK SPACE MONITOR"
echo "=====|
echo "Disk Usage   : ${DISK_USAGE}%"
echo "Threshold    : ${THRESHOLD}%"
echo "-----|

if [[ $DISK_USAGE -gt $THRESHOLD ]]; then
    echo "Status      : ⚠ ALERT - DISK HIGH!"
    echo "Action       : Email sent to $EMAIL"
    mail -s "Disk Alert" $EMAIL <<< "Disk usage is ${DISK_USAGE}%"
else
    echo "Status      : ✓ NORMAL - No Alert"
    echo "Action       : No email sent"
fi

echo "=====|"
```

```
[root@ip-172-31-18-95 ec2-user]# vi diskspace.sh
[root@ip-172-31-18-95 ec2-user]# chmod 755 diskspace.sh
[root@ip-172-31-18-95 ec2-user]# ./diskspace.sh

=====|
    DISK SPACE MONITOR
=====|
Disk Usage   : 26%
Threshold    : 80%
-----|
Status      : ✓ NORMAL - No Alert
Action       : No email sent
=====|
```

12. Bash script to monitor memory and if it is more than 80% then send email notification.

```
#!/bin/bash

THRESHOLD=80
EMAIL="admin@example.com"

TOTAL=$(free -m | awk 'NR==2 {print $2}')
USED=$(free -m | awk 'NR==2 {print $3}')
FREE=$(free -m | awk 'NR==2 {print $4}')
MEM_USAGE=$(echo "scale=1; $USED * 100 / $TOTAL" | bc)

echo "=====
echo "      MEMORY MONITOR
echo "=====
echo "Total   : ${TOTAL} MB"
echo "Used    : ${USED} MB"
echo "Free     : ${FREE} MB"
echo "Usage    : ${MEM_USAGE}%"
echo "Threshold : ${THRESHOLD}%"
echo "-----"

if (( $(echo "$MEM_USAGE > $THRESHOLD" | bc -l) )); then
    echo "Status   : ⚠ ALERT - MEMORY HIGH!"
    echo "Action   : Email sent to $EMAIL"
    mail -s "Memory Alert" $EMAIL <<< "Memory usage is ${MEM_USAGE}%"
else
    echo "Status   : ✓ NORMAL - No Alert"
    echo "Action   : No email sent"
fi

[root@ip-172-31-18-95 ec2-user]# vi mo_mem.sh
[root@ip-172-31-18-95 ec2-user]# chmod 755 mo_mem.sh
[root@ip-172-31-18-95 ec2-user]# ./mo_mem.sh
=====
      MEMORY MONITOR
=====
Total   : 912 MB
Used    : 156 MB
Free     : 519 MB
Usage    : 17.1%
Threshold : 80%
-----
Status   : ✓ NORMAL - No Alert
Action   : No email sent
```

13.Crontab

Create a file named crontab.txt :

vi crontab.txt

```
#purpose Crontab
#owner : Srihari

# 1. April 5th at midnight
0 0 5 4 * echo "April 5th midnight"

# 2. 5th of Jan, Jun, Nov - only if Thursday
0 0 5 1,6,11 4 echo "5th of Jan/Jun/Nov and Thursday"

# 3. At minutes 05 and 27 of hours 9, 10, 11
5,27 9,10,11 * * * echo "Minutes 05 and 27 of 9-11"

# 4. 34th minute of 9AM on August 15th
34 9 15 8 * echo "August 15th 9:34 AM"

# 5. Every midnight
0 0 * * * echo "Midnight every day"

# 6. Every Saturday at 11:59 PM
59 23 * * 6 echo "Saturday night 11:59 PM"

# 7. After every reboot
@reboot echo "System rebooted"
```

Install Cron:

sudo yum install cronie -y

```
[root@ip-172-31-18-95 ec2-user]# sudo yum install cronie -y
Last metadata expiration check: 6:11:53 ago on Sun Jul  5 09:05:55 2026.
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
cronie	x86_64	1.5.7-1.amzn2023.0.2	amazonlinux	115 k
Installing dependencies:				
cronie-anacron	x86_64	1.5.7-1.amzn2023.0.2	amazonlinux	32 k

Step 3: Start the Cron Service

sudo systemctl start crond

Enable it at boot:

sudo systemctl enable crond

Check its status:

sudo systemctl status crond

output:

Active: active (running)

```
[root@ip-172-31-18-95 ec2-user]# sudo systemctl start crond
[root@ip-172-31-18-95 ec2-user]# sudo systemctl enable crond
[root@ip-172-31-18-95 ec2-user]# sudo systemctl status crond
● crond.service - Command Scheduler
   Loaded: loaded (/usr/lib/systemd/system/crond.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-07-05 15:18:28 UTC; 16s ago
```

Step 4: Verify crontab is Available:

```
crontab -l
```

If you haven't created one yet, you'll see:

no crontab for root

This is normal.

Step 5: Install Your Cron File

```
crontab crontab.txt
```

Verify:

```
crontab -l
```

```
[root@ip-172-31-18-95 ec2-user]# crontab crontab.txt
[root@ip-172-31-18-95 ec2-user]# crontab -l
#purpose Crontab
#owner : Srihari

# 1. April 5th at midnight
0 0 5 4 * echo "April 5th midnight"

# 2. 5th of Jan, Jun, Nov - only if Thursday
0 0 5 1,6,11 4 echo "5th of Jan/Jun/Nov and Thursday"

# 3. At minutes 05 and 27 of hours 9, 10, 11
5,27 9,10,11 * * * echo "Minutes 05 and 27 of 9-11"

# 4. 34th minute of 9AM on August 15th
34 9 15 8 * echo "August 15th 9:34 AM"

# 5. Every midnight
0 0 * * * echo "Midnight every day"

# 6. Every Saturday at 11:59 PM
59 23 * * 6 echo "Saturday night 11:59 PM"

# 7. After every reboot
@reboot echo "System rebooted"
```